



DECKMA
HAMBURG

CALIBRATION CHECK KIT

For MEPC.107(49) approved units
(OMD-2005, OMD-2008, OMD-24)

&

MEPC.60(33) approved units
(OMD-11, OMD-21, OCD 1/M, OCD 2/M)



Calibration Check Procedure

- Calibration Check Kit
- Calibration Check Procedure - Step-By-Step
- MSDS

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Content

Revision	2
Usage of this instruction manual	3
Symbols	3
Introduction	4
1. Calibration Check	4
1.1 Units approved according MEPC.107(49)	4
1.2 Units approved according MEPC.60(33)	5
2. Package Contents	5
3. Dilution of “Calibration Check Fluid”	5
4. Function Test: step by step	6
5. Material Safety Data Sheet MSDS	9
Sales & Service Locations	12
Notes:	13

Revision

Document-Name	Revision No.	Notation	Date
Calibration Check Procedure For MEPC.107(49) & MEPC.60(33) approved units	R02_20181106		06.11.2018



Usage of this instruction manual

This instruction manual will guide you to apply the calibration check correctly. The instruction manual is split into sections. The components are listed below. Also, the calibration check is explained step-by-step.

This instruction manual contains also information about the service support and our contact details. Additionally, the symbols below will support you during the function test.

Symbols



This symbol is used for giving general advices.



This symbol is used for giving advices about servicing.



Introduction

The method of on site calibration check is to use the new highly accurate developed DECKMA “Calibration Check Fluid” as a reference standard for 5 ppm and 15 ppm Deckma Hamburg Units.

The DECKMA “Calibration Check Fluid” contains a well-defined mixture of polymer-particles suspended in water and is colored yellow.

The highly accurate suspension has to be diluted for calibration check for MEPC.107(49) approved units (OMD-2005, OMD-2008, OMD-24) as well as for MEPC.60(33) approved units (OMD-11, OMD-21, OCD-1/M, OCD-2/M).

The Calibration Check Fluid is long-term stable, an expiry date is printed onto the bottle of the Calibration Check Fluid.

1. Calibration Check

For a calibration check on site it is necessary, to have a definite fluid available which simulates oil content in water. With the OMD-2005, OMD-2008 and OMD-24 the “Calibration Check Fluid” in water solution with the concentration of e.g. 5 ppm or 15 ppm can be used. For OMD-11, OMD-21, OCD-1/M, OCD-2/M the Calibration Check Fluid in water solution with concentration typically 15 ppm is to be used.



Important Note:

The calibration check does not replace the accuracy check requested by IMO, by authorized technicians from the manufacturer or representative, associated with a calibration certificate.

1.1 Units approved according MEPC.107(49)

4.2.11 The validity of calibration certificates should be checked at IOPP annual/intermediate/renewal surveys. The accuracy of 15 ppm bilge alarms is to be checked by calibration and testing of the equipment conducted by a manufacturer or persons authorized by the manufacturer and should be done at intervals not exceeding five years after its commissioning, or within the term specified in the manufacturer's instructions, whichever is shorter. Alternatively the unit may be replaced by a calibrated 15 ppm bilge alarm. The calibration certificate for the 15 ppm bilge alarm, certifying the date of the last calibration check, should be retained on board for inspection purposes.



1.2 Units approved according MEPC.60(33)

4.2.8 It is recommended that a simple means be provided aboard ship to check on instrument drift, and to confirm the accuracy and repeatability of the instrument reading.

2. Package Contents

No.	Description	
1.	Calibration Check Kit: Order No.: 18515 Consists of: A. 1 off bottle with 5 ml "Calibration Check Fluid" (yellow) B. 7 off rods C. 1 off graduated beaker 100 ml D. Instruction Manual incl. MSDS Material Safety Data Sheet.	

3. Dilution of "Calibration Check Fluid"



Dilute the "Calibration Check Fluid" to 100 ml water suitable for drinking according the requirements. Please refer to the dilution table for a correct dilution. For safety precautions read the material safety data sheet MSDS as well. For the dilution procedure see step-by-step handling below.

Amount of "Calibration Check Fluid" in **100 ml** water

Calibration Check Fluid	Reading Units approved according MEPC.107(49) OMD-2005, OMD-2008, OMD-24	Due to difference in storage time and minor changes in mixing the reading may differ up to ± 5 ppm. In such cases the concentration can be modified by the operator to reach the alarm set point.
1 drop	1 ppm	
5 drops	5 ppm	
15 drops	15 ppm	

Table 1 Dilution Table for units approved accord. MEPC.107(49)

In case of units approved according MEPC.60(33), dilute the "Calibration Check Fluid" according the following dilution table for calibration purposes, e.g. OCD-1 ff, OMD-1x, OMD-21.

Calibration Check Fluid	Expected suspension Formazin according manual FTU	Reading		
		OCD-1 ff	OMD-11, -12, -15, -17	OMD-21
40 drops	200	40 ppm	32 ppm	24 ppm
30 drops	150	30 ppm	24 ppm	18 ppm
20 drops	100	20 ppm	16 ppm	12 ppm
10 drops	50	10 ppm	8 ppm	6 ppm

Table 2 Dilution Table for units approved accord. MEPC.60(33)

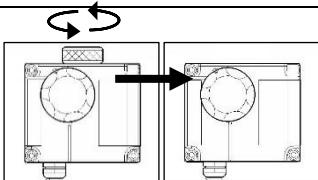
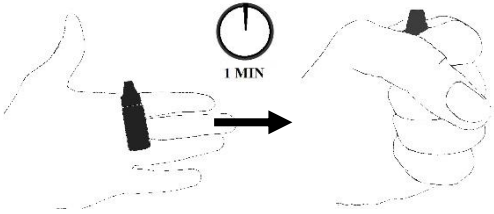
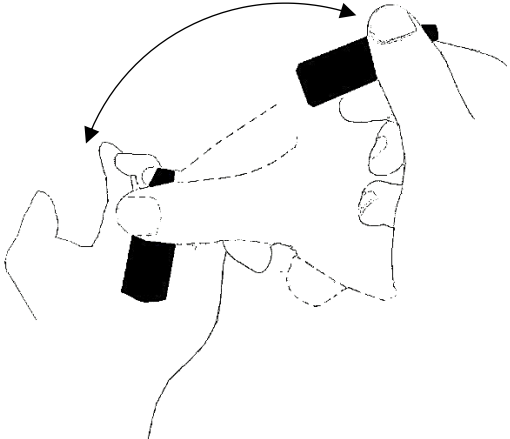


4. Function Test: step by step



Please note:

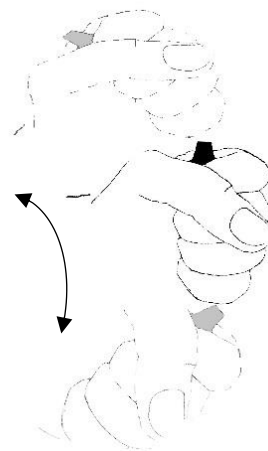
Shake the “Calibration Check Fluid” well **before** using and read the material safety data sheet as well.

STEP 1. Run the system with clear oil free water and check the display showing 0-1 ppm. If not, clean proper the sample tube according the “SERVICING AND CLEANING MANUAL”. Be sure that the Offset is set to zero, see corresponding instruction manual.	
STEP 2. Stop any water flow, open the cell cap.	
STEP 3. Fill the graduated beaker with 100 ml water. It should be potable water quality, in minimum.	
STEP 4. Warm the bottle in your hand (for a minute).	
STEP 5. Hit the bottle several times (e.g. 10x) on the palm of your hand with a hand-striking movement.	

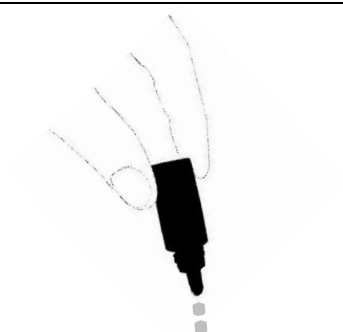


STEP 6.

And, **Shake** the “Calibration Check Fluid” well.



Open the bottle and discard the first two drops.



STEP 7.

Prepare the required concentration of the fluid according *Table 1 Dilution Table*.

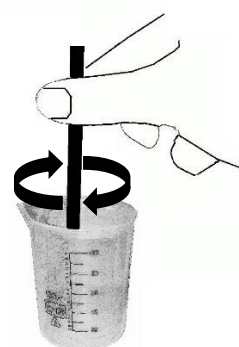


Step 8.

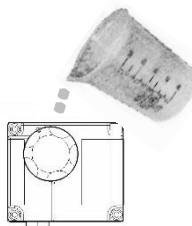
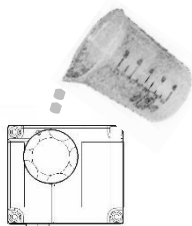
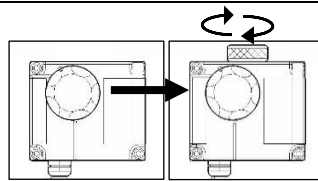
Stir the Calibration Fluid with the supplied rod, to get a homogeneous dilution.



This dilution is durable only for about an hour.





STEP 9. Empty the sample tube and fill it twice (first for flushing the oil free water) with the diluted solution.	
STEP 10. Wait about a minute, to get rid of any air bubbles/ foam from stirring the dilution.	
STEP 11. Again, empty the sample tube and fill it with the diluted solution.	
STEP 12. Wait about a minute, to get rid of any air bubbles/ foam from stirring the dilution.	
STEP 13. Check the reading for the next 2 minutes – it has to be in the intended range (see <i>Table 1 Dilution Table</i>). Use the data log system for check.	
STEP 14. If the reading is more than 5 ppm out of range check all items and try again: reading with oil free water (cleaning with a bottle brush), offset-setting, dilution of “Calibration Check Fluid”.	
STEP 15. Close the cell cap and flush with oil free water.	
STEP 16. The instrument is now ready for use.	



5. Material Safety Data Sheet MSDS

SECTION 1: Product Identification / Company Identification

Product Name: Deckma Hamburg Calibration Check Kit
Contents: Polymer particles suspended in water. The particles contain polymers consisting of polystyrene or poly-(styrene-co-divinylbenzene)
Company Name: Deckma Hamburg GmbH
Address: Kieler Straße 316 22525 Hamburg, Germany
Phone Number: +49 (0) 40 54 88 76-0
Fax Number: +49 (0) 40 54 88 76-10
Email: post@deckma.com

SECTION 2: Hazard Identification

Caution: May cause lung irritation if inhaled.

If used as typically applied the product is not regarded as hazardous to health. The product is not classified as flammable or harmful to the environment. It may cause skin, eye, digestive track and respiratory track irritation. Toxicological properties have not been fully investigated. The product is not classified as dangerous in accordance with Directive 1907/2006 (REACH) or 1272/2008 (CLP).

SECTION 3: Composition / Information on Ingredients

	CAS No	Formula	Percentage	EINECS/ELINCS
Solid polymer beads made of polystyrene in aqueous suspension	9003-53-6	$(C_8H_8)_x$	0.01 - 4	N/A
Solid polymer beads made of poly(styrene-divinylbenzene) in aqueous suspension	9003-70-7	$(C_{10}H_{10})_x$	0.01 - 1	N/A
Water	7732-18-5	H ₂ O	95 - 99.99	231-791-2
dispersant/surfactant	Proprietary	Proprietary	<0.1	N/A
preservative	Proprietary	Proprietary	<0.05	N/A
dyes	Proprietary	Proprietary	<0.01	N/A
Note: This MSDS addresses potential worker health and safety issues associated with the handling of the final product (i.e., during its transportation, distribution, and use).				

SECTION 4: First Aid Measures

Eye: Wash thoroughly with water for at least 10 minutes. If an irritation develops, seek medical advice.
Skin: Wash thoroughly with soap and rinse with water. If an irritation develops, seek medical advice.
Inhalation: Remove to fresh air and seek medical advice.
Ingestion: Give two glasses of water and seek medical advice.

SECTION 5: Fire Protection

Flammability: None.
Explosivity: None.
NFPA/HMIS Rating (estimated): Health: 1;
Flammability: 0;
Instability: 0
Fire Fighting Instructions: Use extinguishing media appropriate for surrounding fire (water, foam, vaporizing liquid or multipurpose dry chemical).

SECTION 6: Accidental Spill and Release Measures

Areas covered with spilled product may be slippery. If product is spilled or released, keep away other people from this area. Collect material by wiping the spilled product with a paper towel or disposable wipe and discard it into an appropriate container suitable for proper disposal. Avoid inhaling fine particle dust.



SECTION 7: Handling and Storage

Keep bottles tightly sealed to prevent contamination. Store at room temperature (8°C-15°C) in a dry area. Avoid damaging or puncturing containers. When creating aerosols of fine particles, use weak directional airflow (away from the user).

SECTION 8: Exposure Control / Personal Protection

Occupational Exposure Limits

Chemical Name:	Polystyrene or Poly-(Styrene-co-divinylbenzene), water
Exposure Guidelines:	The product does not contain any hazardous material with occupational exposure limits.
Eye protection:	Safety glasses with side shields are suggested.
Skin protection:	Powder-free latex or vinyl gloves are suggested to avoid skin exposure.
Respiratory protection:	Respiratory protection program that meets OSHA's 29CFR1910.134 and ANSI Z88.2 requirements or European Standard EN 149.
Engineering controls:	Use weak directional airflow (away from the user).
General Hygiene Considerations:	Wear appropriate protective cloths to avoid skin exposure.

SECTION 9: Physical / Chemical Properties

Boiling point:	100 °C (as water)
Solubility in water:	Particles are insoluble
Melting point:	0 °C (as water)
Percent volatile:	Negligible
Physical State:	Aqueous suspension
Vapor pressure:	Negligible
Appearance, color:	Fluorescent yellow
pH:	5 - 8
Flash Point:	N/A
Flammability:	N/A
Auto Ignition Temp.:	N/A
Decomposition:	N/A
Flash Point:	N/A
Particle/Suspension Density:	1,0 - 1.05 g/cm ³

SECTION 10: Stability and Reactivity

Hazardous decomposition products:	Not known
Incompatibility:	Not known
Hazardous polymerizations:	Will not take place
Stability:	Stable at room temperature under normal handling and storage conditions

SECTION 11: Toxicological Information

Carcinogenicity:	Material is not listed by NTP, ACGIH, IARC or OSHA as carcinogen. Please also refer to SECT. 2
Reproductive and Developmental Toxicity:	Material is not considered to be a reproductive or developmental toxicant

SECTION 12: Ecological Information

Ecotoxicity:	Not considered as dangerous for the environment LC/EC/IC50 > 100 mg/L
Persistence/Degradation:	No information found
(Bio)accumulation:	No information found
Mobility in Environmental Media:	Particles are insoluble in water
Other Adverse Effects:	No information found

SECTION 13: Disposal

All wastes containing the product should be specially contained, properly labelled, and stored separately from other facility waste discharges. Dispose of any waste residues according to prescribed federal, state, and local guide-lines (e.g., to an appropriately permitted chemical waste incinerator). None of the components are on RCRA P&RCRA U Series Wastes list. EWC waste code 180105 discard as chemicals and medicines. According to the European Waste Catalogue, Waste Codes are



not product specific, but application specific. Waste codes should be assigned by the user based on the application for which the product was used.

SECTION 14: Transportation Information

Hazardous Class: Not classified per ADN, ADR, IATA, ICAO, IMDG/IMO, OTIF, RID, TDG, UNECE, U.S. DOT

SECTION 15: Regulatory Information

OSHA PEL					ACGIH TLV				OSHA PEL		ACGIH TLV			
Ceiling	TWA	STEL	Final	Vacated	Ceiling	TWA	STEL	OEL	WEEL	OEL	OEL	WEEL	OEL	IDLH
Not listed	Not listed	Not listed	Not listed	Not listed	Not listed	Not listed	Not listed	Not listed	Not listed	Not listed	Not listed	Not listed	Not listed	Not listed

US State: CAS #9003-53-6 - Texas
California: Prop 65, Title 8 CCR, Title 22 CCR, ATHS, Occupational Exposure Limits, Ozone Depleting Compounds, Precursor Chemicals, SCAQMD, California No Significant Risk Level - No information found
None of the chemicals are on Prop 65 lists
CAS #9003-70-7
WHMIS: This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR). The MSDS contains all the information required by the CPR
Date of issue: Mainz, March 23rd of 2018

Abbreviations

ACGIH: American Conference of Governmental Industrial Hygienists
EU: European Union
NIOSH: National Institute of Occupational Safety and Health
DOT: Department of Transportation
IARC: International Agency for Research on Cancer
NTP: National Toxicology Program
OSHA: Occupational Safety and Health Administration
IATA: Intern. Air Transp. Association

The above information is based on data available to us and is believed to be correct. Since the information may be applied under conditions beyond our control and with which we may be unfamiliar, we do not assume any responsibility for the results of its use and all persons receiving it must make their own determination of the effects, properties and protections which pertain to their particular conditions. No representation, warranty, or guarantee, express or implied (including a warranty of fitness or merchantability for a particular purpose), is made with respect to the materials, the accuracy of information, the results to be obtained from the use thereof, or the hazards connected with the use of the material. Caution should be used in the handling and use of the material.



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<http://www.deckma.com/agents>

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